

1 Invertible matrices

Definition 1.1. An $n \times n$ matrix A is *invertible* if there is a matrix B with $AB = BA = I_n$.

Theorem 1.2 (Invertible matrix theorem). *For an $n \times n$ real matrix A the following are equivalent:*

- (i) A is invertible;
- (ii) the equation $Ax = 0$ has only the solution $x = 0$;
- (iii) $\text{rank } A = n$;
- (iv) $\det A \neq 0$.

Proof. We prove a cycle of implications.

(i) $\Rightarrow$ (ii). If $Ax = 0$ then $x = A^{-1}Ax = 0$.

(ii) $\Rightarrow$ (iii). The kernel is trivial, so by rank–nullity $\text{rank } A = n - \dim \ker A = n$.

(iii) $\Rightarrow$ (iv). Row reduction reaches I_n using elementary matrices $E_1, \dots, E_k$, and $\det(E_k \cdots E_1 A) = 1$ forces $\det A \neq 0$.

(iv) $\Rightarrow$ (i). The adjugate formula $A^{-1} = \frac{1}{\det A} \text{adj } A$ gives the inverse. $\square$

Proposition 1.3. *For 2×2 matrices,*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{whenever } ad - bc \neq 0.$$

Proof. Multiply out:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}. \quad \square$$

2 Trace

Proposition 2.1. *For all $A, B \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}$:*

- $\text{tr}(A + B) = \text{tr } A + \text{tr } B$ and $\text{tr}(cA) = c \text{tr } A$;
- $\text{tr}(AB) = \text{tr}(BA)$;
- $\text{tr}(P^{-1}AP) = \text{tr } A$ for every invertible P .

Proof. Linearity is immediate from the definition. For the second item,

$$\text{tr}(AB) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^n b_{ji} a_{ij} = \text{tr}(BA).$$

The third item follows from the second with A replaced by AP and B by P^{-1} . $\square$

Remark 2.2. By Proposition 2.1 the trace is an invariant of the linear map, not only of the matrix, and Theorem 1.2 shows the same for invertibility.